October 25, 2026

Cover Letter — The Journal of Supercomputing

Dear Editor-in-Chief,

We submit our manuscript "Distilling Frozen Time-Series Foundation Model Teachers into Lightweight Edge Anomaly Detectors: A Preregistered Quality–Fidelity–Cost Study" for consideration in *The Journal of Supercomputing*.

Time-series foundation models (TSFMs) such as Chronos-Bolt offer strong zero-shot anomaly detection, but their per-point inference cost (over 100 ms single-threaded in our measurements) is far beyond edge budgets. The standard remedy—distill the frozen teacher into a lightweight student and deploy the student alone—silently assumes the teacher's detection behavior survives the transfer. Our manuscript delineates that boundary and localizes it at the distillation interface:

- A mechanism-level finding, not just a negative verdict: the distillation interface, not distillation itself, decides what transfers. Score-level distillation transfers almost none of the Chronos teacher's ranking behavior and yields no detectable quality advantage (the full-flow TimesFM extension adds an instructive split: its ranking does transfer at 0.70 and still yields none); the teacher's point forecasts as the training target recover most of the gap on development flows and are statistically equivalent to self-supervision on held-out sources. Four controls—shuffled-teacher, equal-context, anomaly-exposure, and a second teacher family—attribute the failure to the distillation interface rather than to student capacity, within the tested configurations.
- A preregistered, source-blocked protocol (frozen design with an append-only, hash-chained revision log; 29 flows from 22 public data collections; 1,270 preregistered runs plus 586 disclosed additions, 1,856 runs in total; eleven arms) comparing score-, ranking-, forecast-, and self-supervised training of compact (≈0.27M-parameter) students against the frozen Chronos-Bolt teacher, TimesFM, and classical references.
- A deployment-cost axis aligned with the journal's scope: measured on trained ONNX artifacts with optional INT8 quantization, students run 0.07–0.51 ms per window (p50)—roughly 200–1500× faster than the teacher—and quantized PatchMLP students fit sub-millisecond, sub-megabyte budgets at every tested channel count (1 to 248), with per-configuration quantization quality drift quantified. On-device validation on a Raspberry Pi 4B (single-thread ONNX Runtime, batch 1) replicates the protocol at the reference dimension: the students measure 0.44–15.47 ms p95 on board, remaining 8–278× below the teacher desktop cost.

We believe this combination—a rigorous boundary delineation of a widely assumed technique, plus concrete edge-deployment measurements—fits the journal's interest in system performance, resource-constrained computing, and dependable empirical evaluation.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and declare no conflicts of interest. All benchmark data are public; code, the preregistration with its hash chain, and per-run results are released at `https://github.com/moonzymoon/tsfm-distillation-artifacts` and archived with the Zenodo DOI `https://doi.org/10.5281/zenodo.23623799`; the complete artifact package accompanies the submission.

Thank you for your consideration.

Sincerely,

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